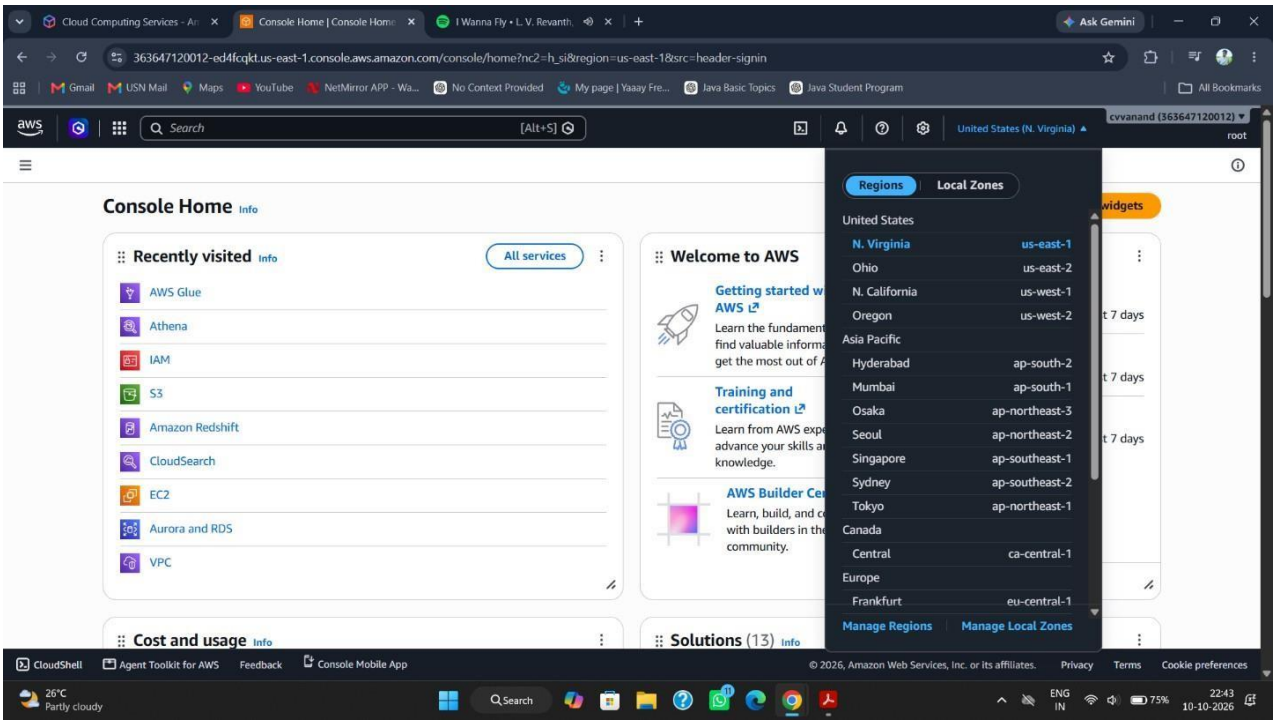
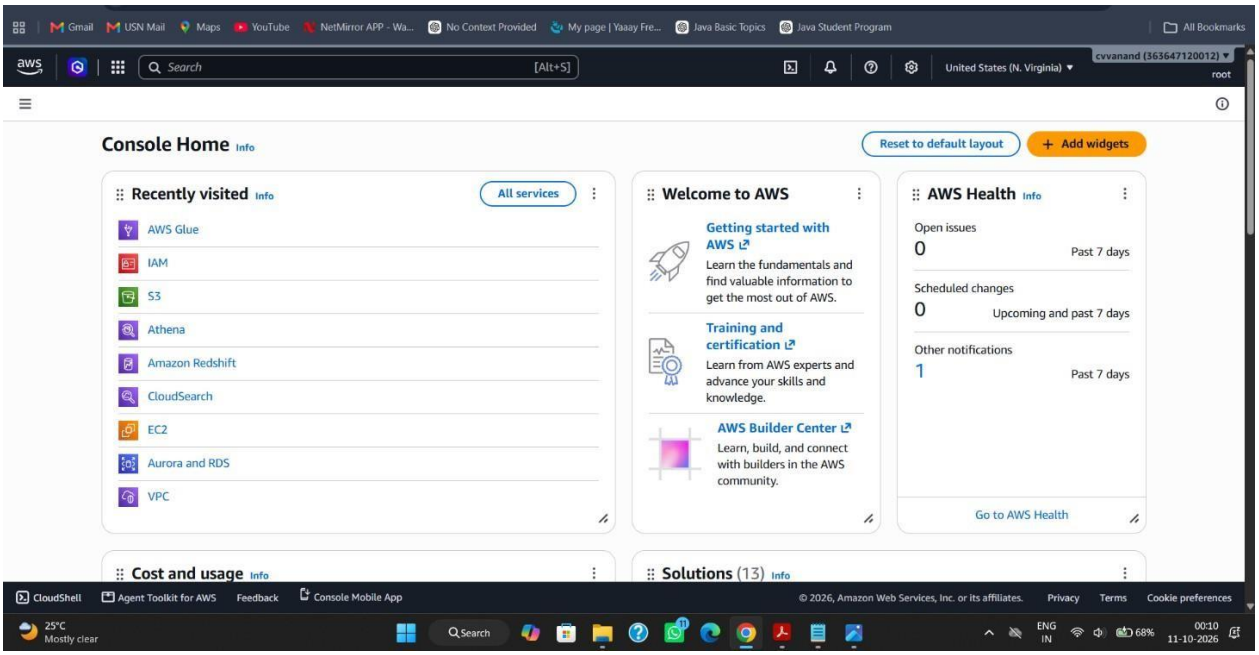
AWS Glue ETL Pipeline with Amazon S3

CVV ANAND,
23BTRCL176,
HCLTFP2297504
AIML-A.

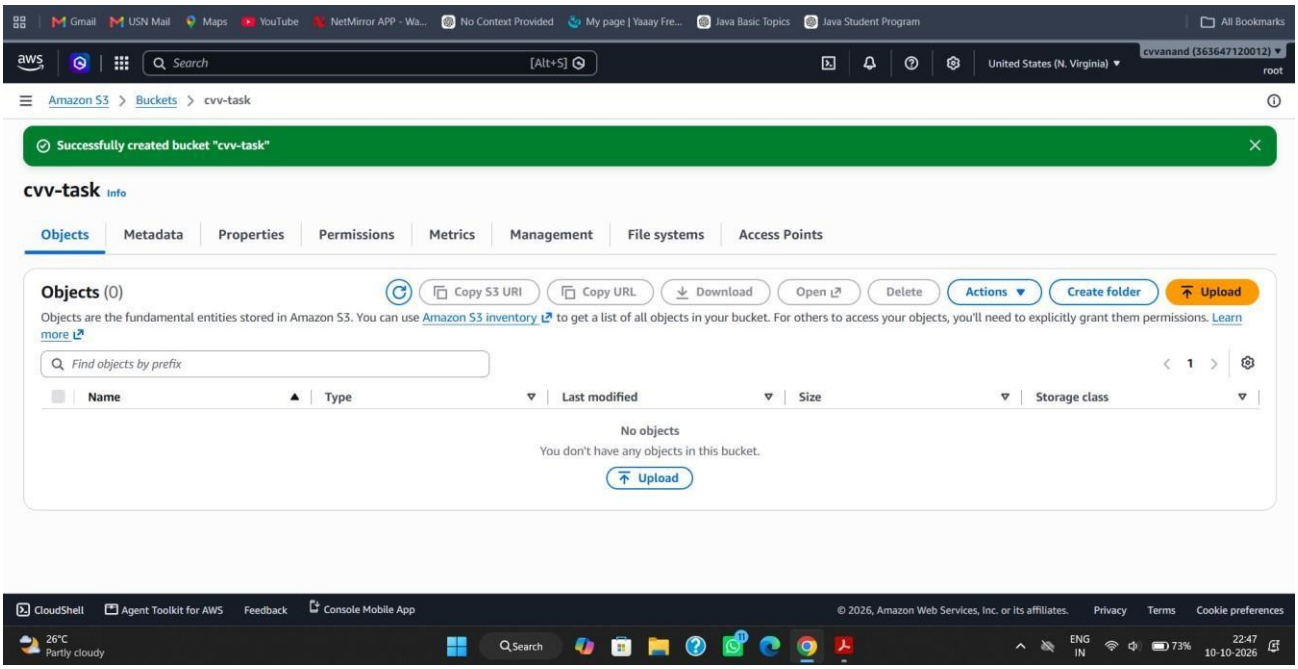
AWS Account / Region: -



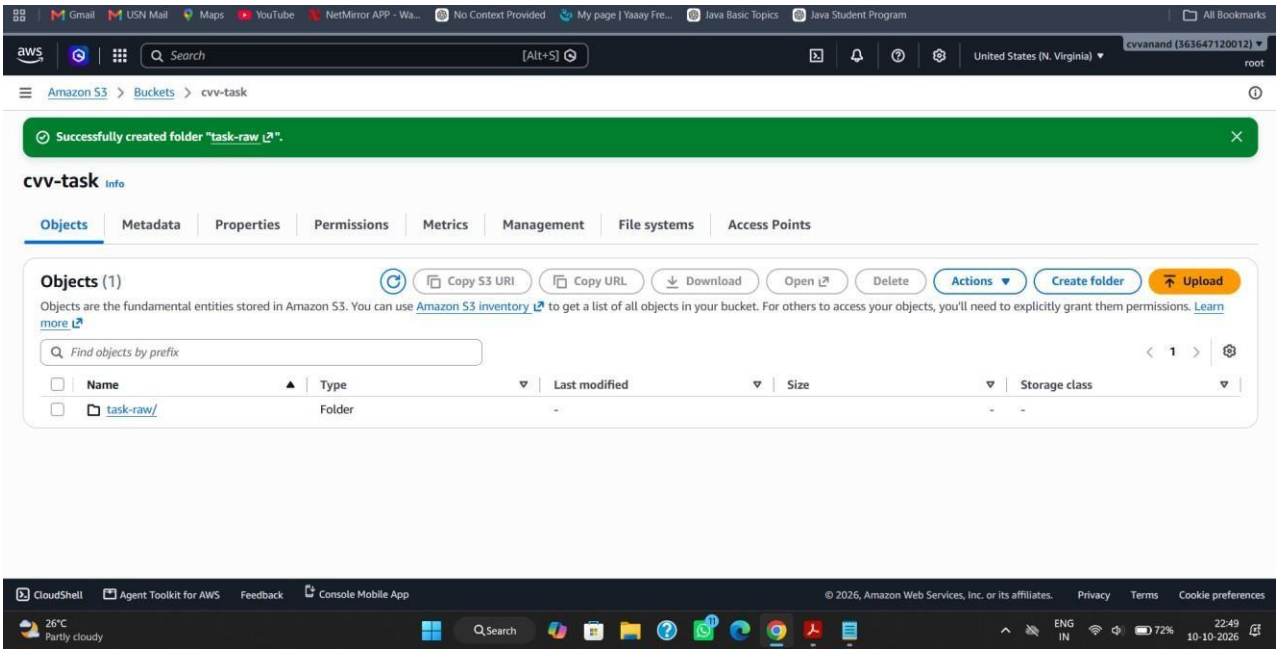

Gmail
USN Mail
Maps
YouTube
NetMirror APP - Wa...
No Context Provided
My page | Yaaay Fre...
Java Basic Topics
Java Student Program
All Bookmarks
Search
[Alt+S]
United States (N. Virginia)
cvvanand (363647120012)
root
Console Home Info
Reset to default layout
+ Add widgets
Recently visited Info
All services
AWS Glue
IAM
S3
Athena
Amazon Redshift
CloudSearch
EC2
Aurora and RDS
VPC
Welcome to AWS
Getting started with AWS
Learn the fundamentals and find valuable information to get the most out of AWS.
Training and certification
Learn from AWS experts and advance your skills and knowledge.
AWS Builder Center
Learn, build, and connect with builders in the AWS community.
AWS Health Info
Open issues
0
Past 7 days
Scheduled changes
0
Upcoming and past 7 days
Other notifications
1
Past 7 days
Go to AWS Health
Cost and usage Info
Solutions (13) Info
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
25°C
Mostly clear
Search
ENG
IN
68%
00:10
11-10-2026

Amazon S3: -

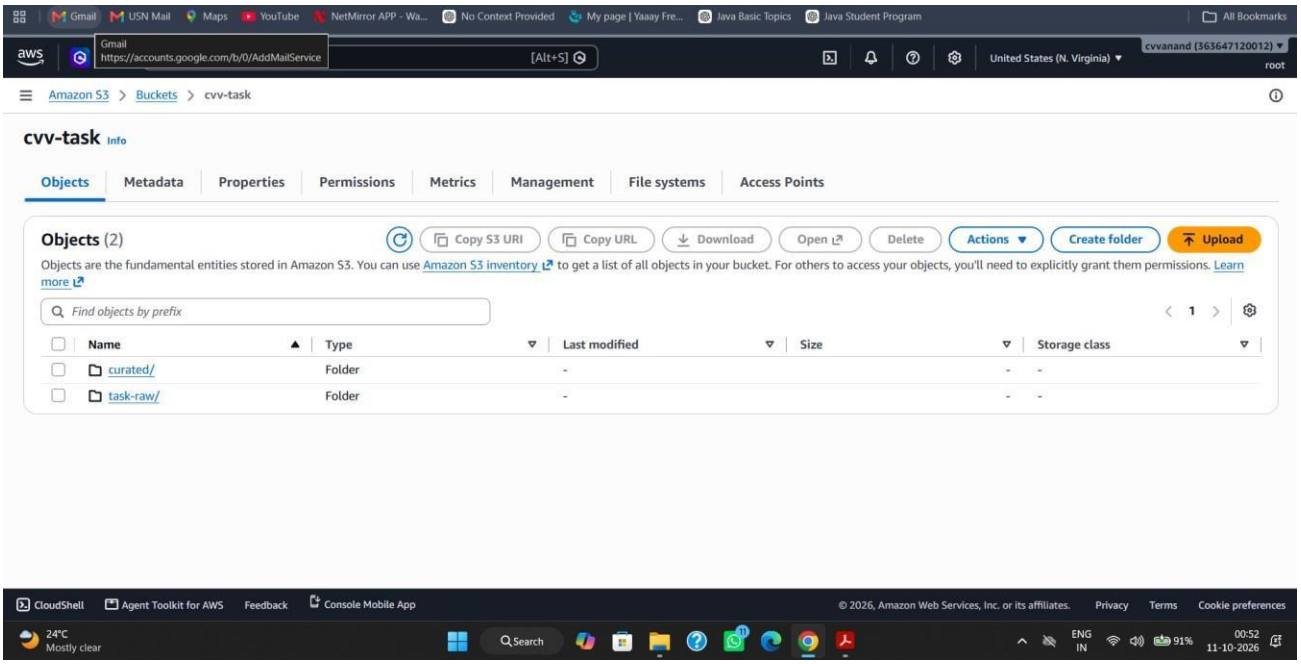
S3 bucket created



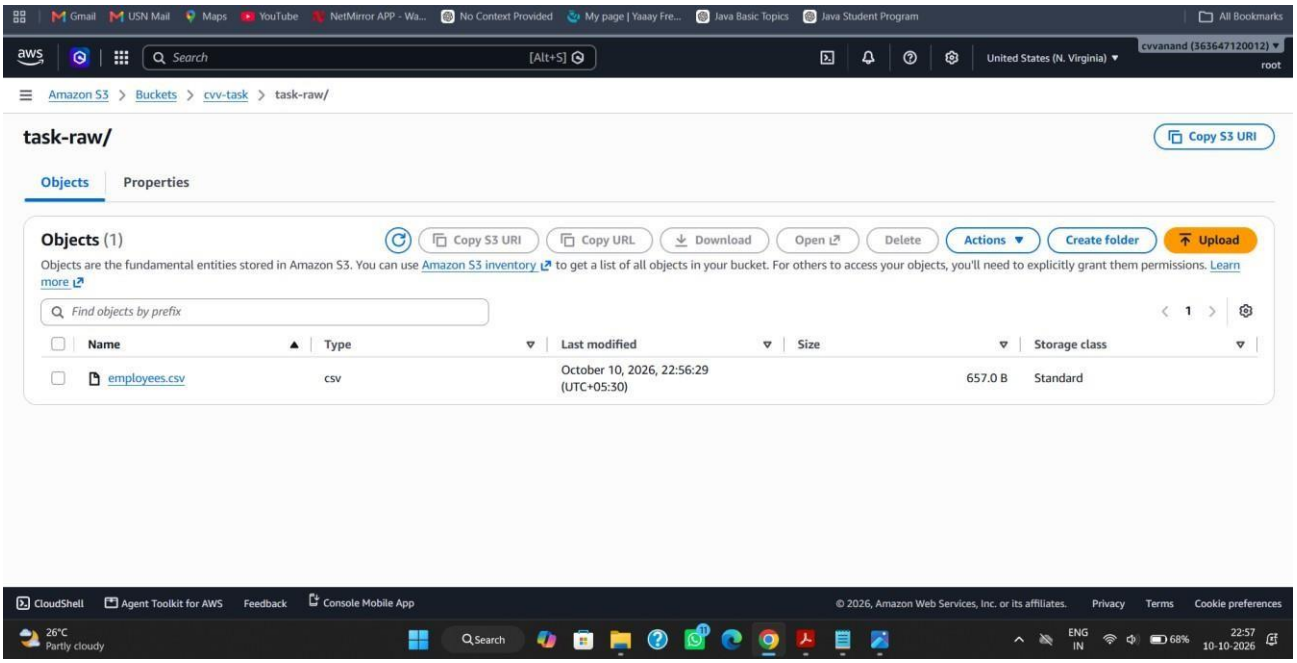
raw/ folder



curated/ folder



employees.csv uploaded inside raw/



Bucket name and file name should be visible

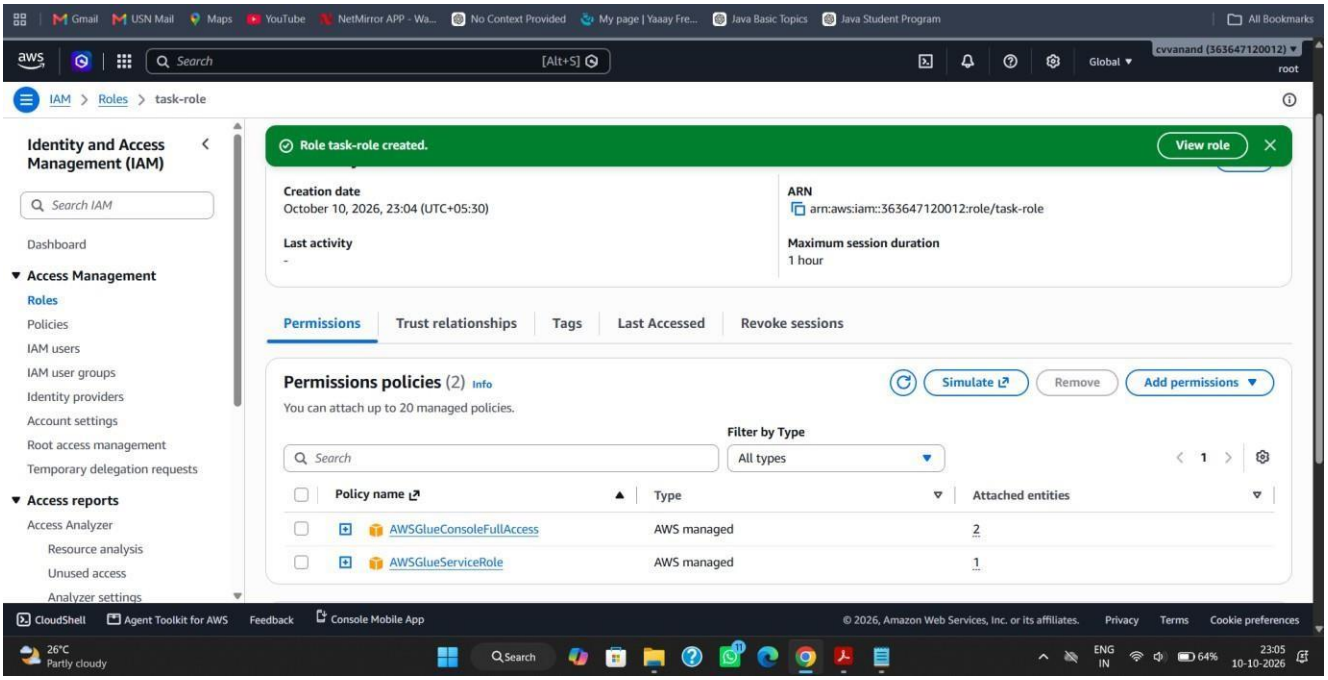
'VISIBLE' in the screenshots.

IAM: -

IAM Role used for AWS Glue

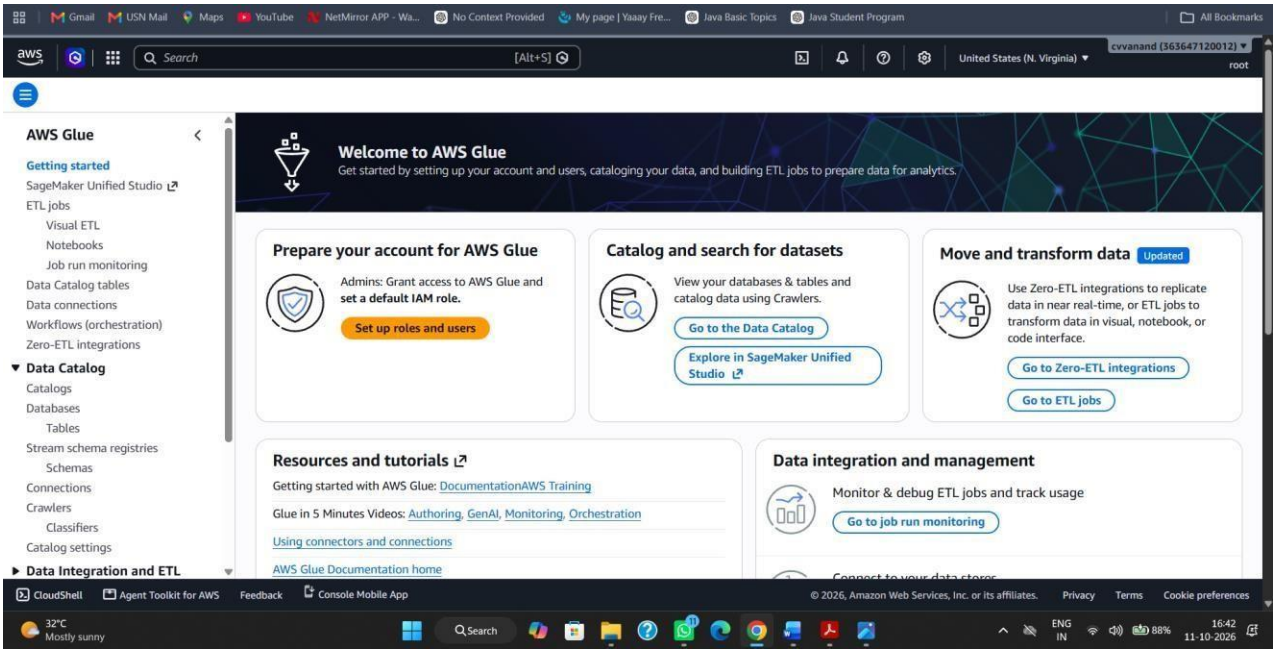
Role name: - ‘task-role’

Relevant permissions/policies

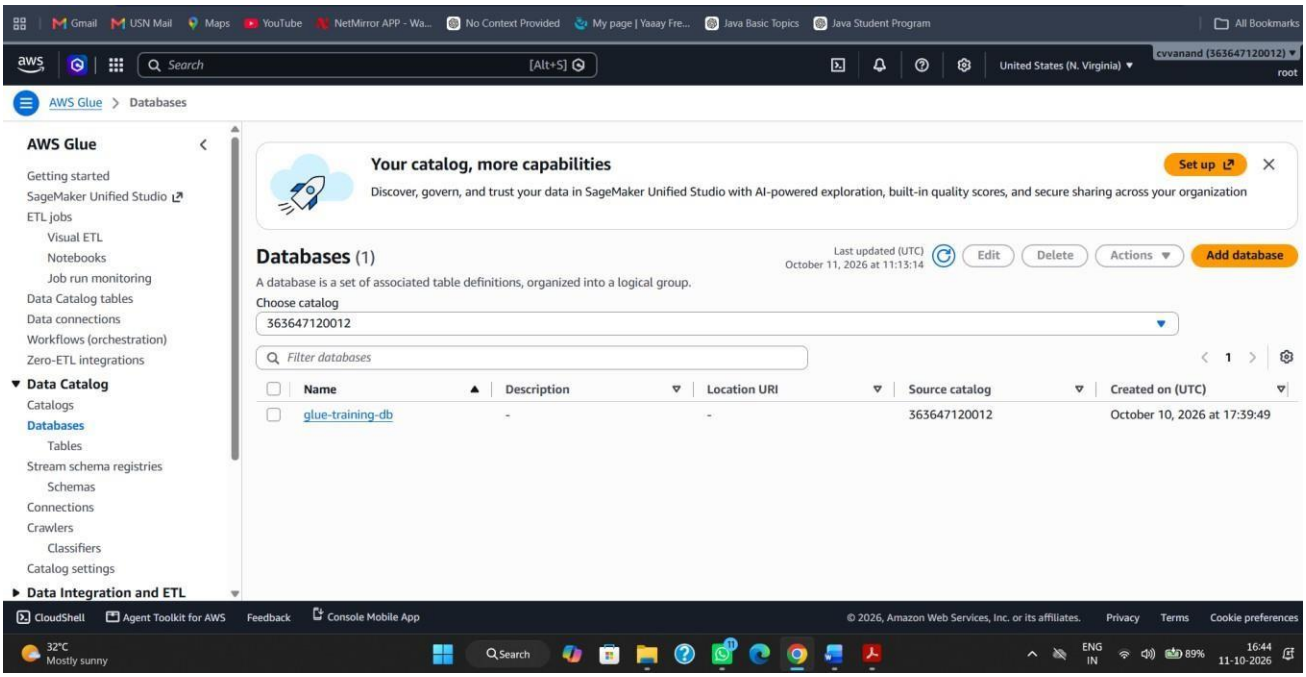


AWS Glue Database: -

AWS Glue Console



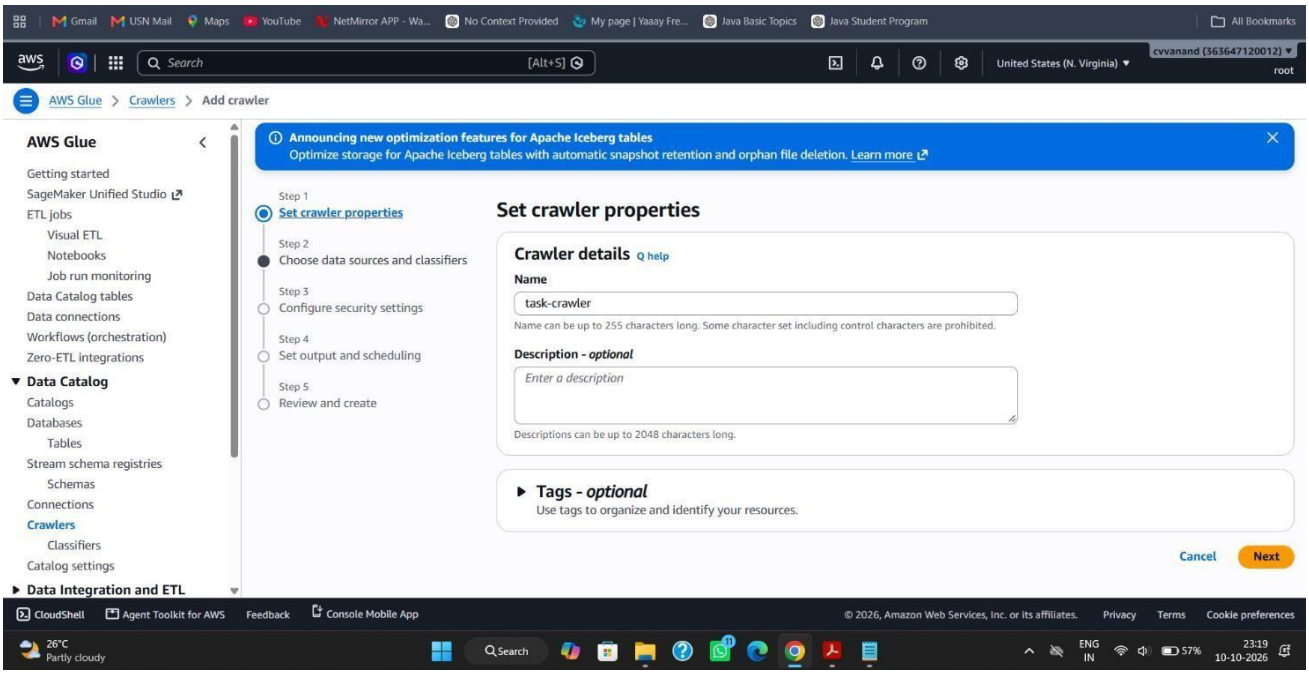
Data Catalog and Database created



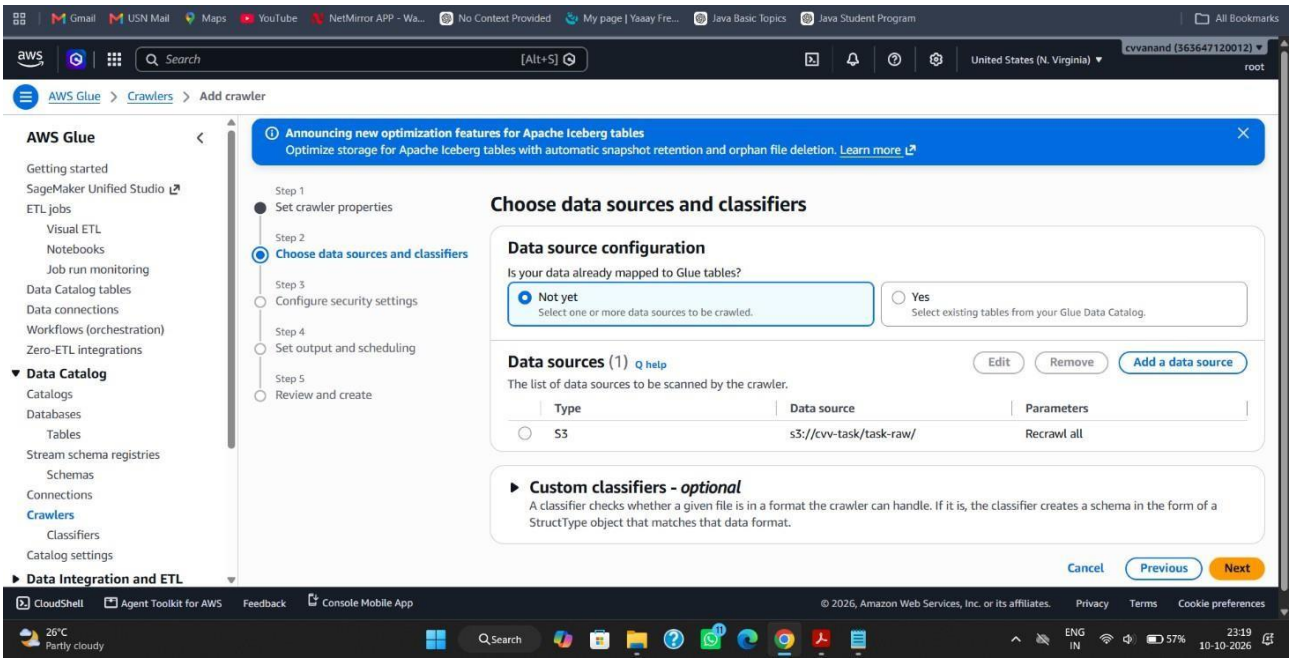
Database name: glue_training_db

Glue Crawler: -

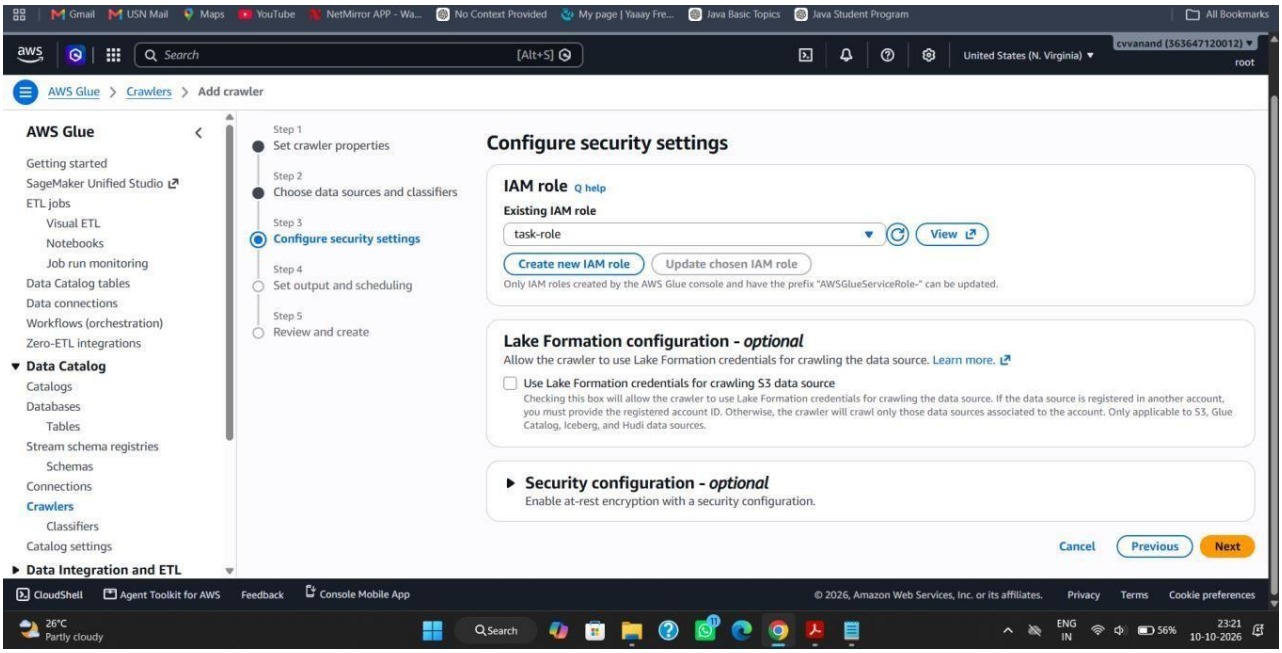
Crawler name



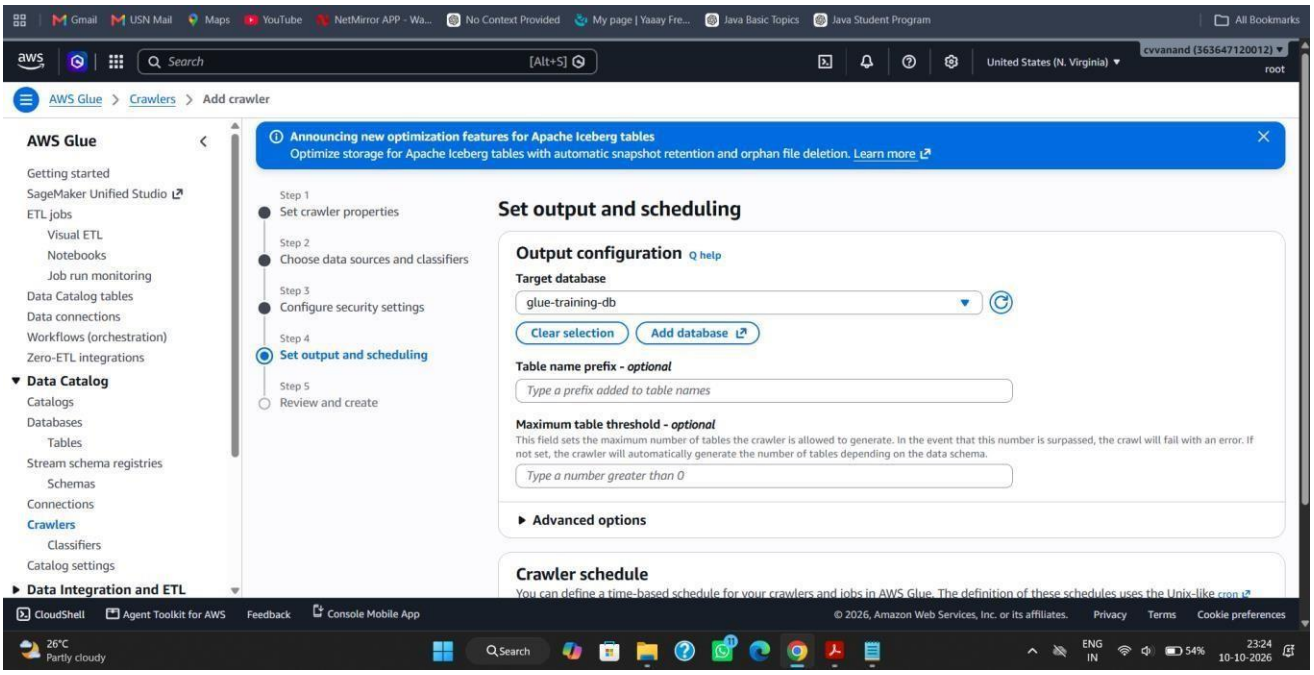
S3 source location


AWS Glue > Crawlers > Add crawler
Announcing new optimization features for Apache Iceberg tables
Optimize storage for Apache Iceberg tables with automatic snapshot retention and orphan file deletion. Learn more
Step 1
Set crawler properties
Step 2
Choose data sources and classifiers
Step 3
Configure security settings
Step 4
Set output and scheduling
Step 5
Review and create
Choose data sources and classifiers
Data source configuration
Is your data already mapped to Glue tables?
Not yet
Select one or more data sources to be crawled.
Yes
Select existing tables from your Glue Data Catalog.
Data sources (1)
Edit
Remove
Add a data source
The list of data sources to be scanned by the crawler.
Type
Data source
Parameters
S3
s3://cvv-task/task-raw/
Recrawl all
Custom classifiers - optional
A classifier checks whether a given file is in a format the crawler can handle. If it is, the classifier creates a schema in the form of a StructType object that matches that data format.
Cancel
Previous
Next

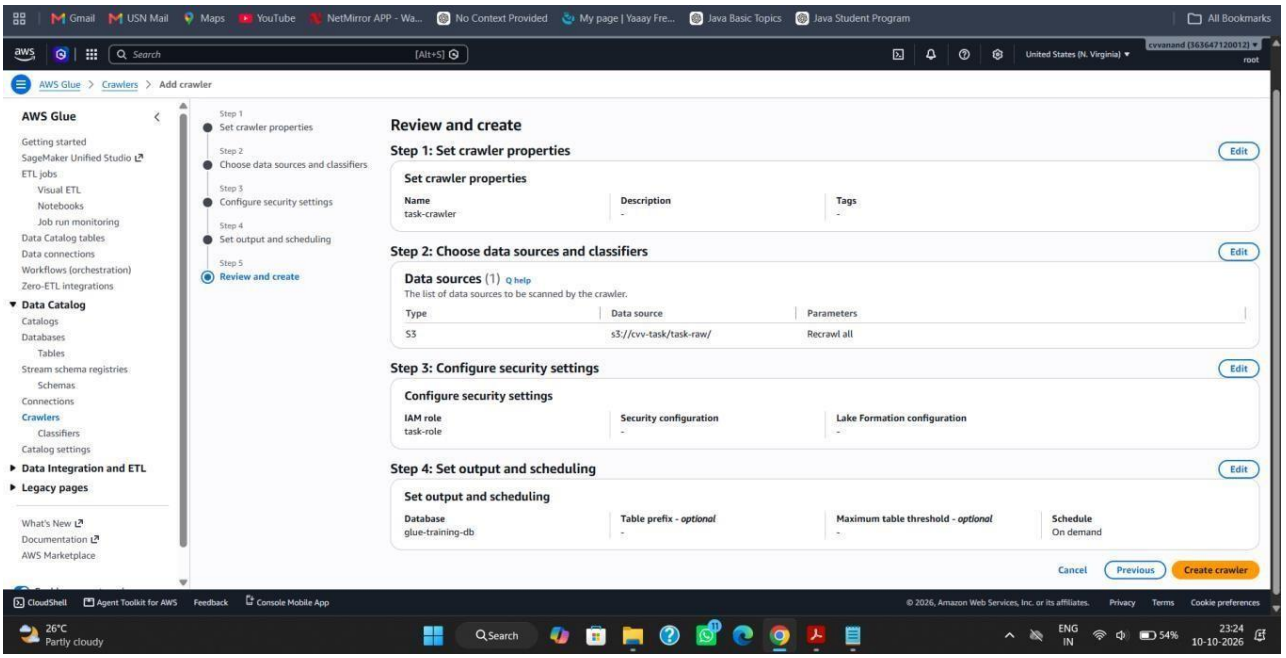
IAM role


AWS Glue > Crawlers > Add crawler
Step 1
Set crawler properties
Step 2
Choose data sources and classifiers
Step 3
Configure security settings
Step 4
Set output and scheduling
Step 5
Review and create
Configure security settings
IAM role
Existing IAM role
task-role
View
Create new IAM role
Update chosen IAM role
Only IAM roles created by the AWS Glue console and have the prefix "AWSGlueServiceRole-" can be updated.
Lake Formation configuration - optional
Allow the crawler to use Lake Formation credentials for crawling the data source. Learn more.
Use Lake Formation credentials for crawling S3 data source
Checking this box will allow the crawler to use Lake Formation credentials for crawling the data source. If the data source is registered in another account, you must provide the registered account ID. Otherwise, the crawler will crawl only those data sources associated to the account. Only applicable to S3, Glue Catalog, Iceberg, and Hudi data sources.
Security configuration - optional
Enable at-rest encryption with a security configuration.
Cancel
Previous
Next

Target database

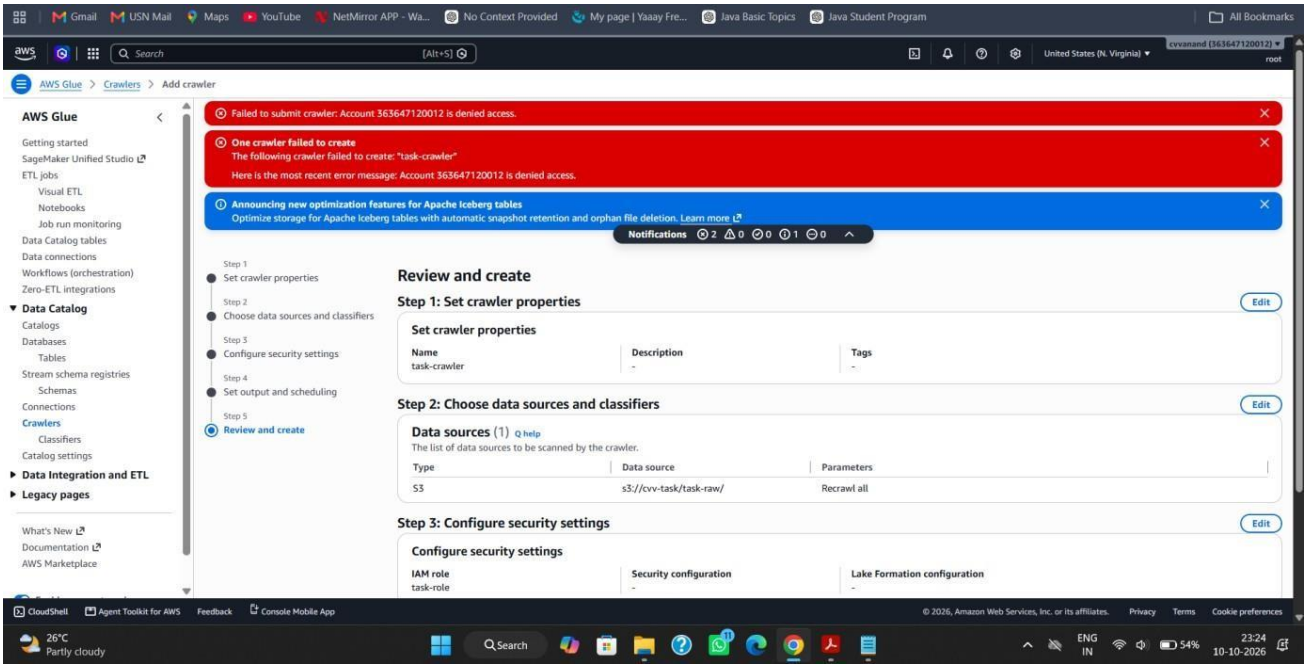


Crawler configuration



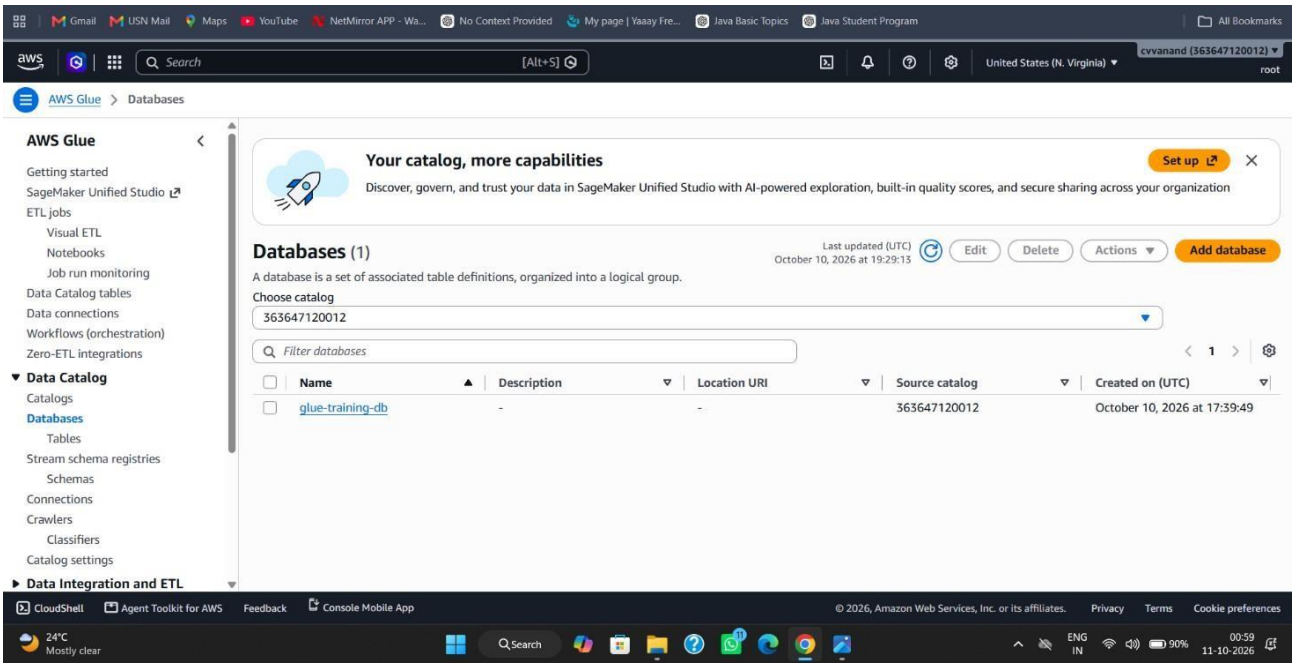
Crawler Execution: -

Crawler successfully completed and Crawler status



Glue Data Catalog: -

Database



employees table

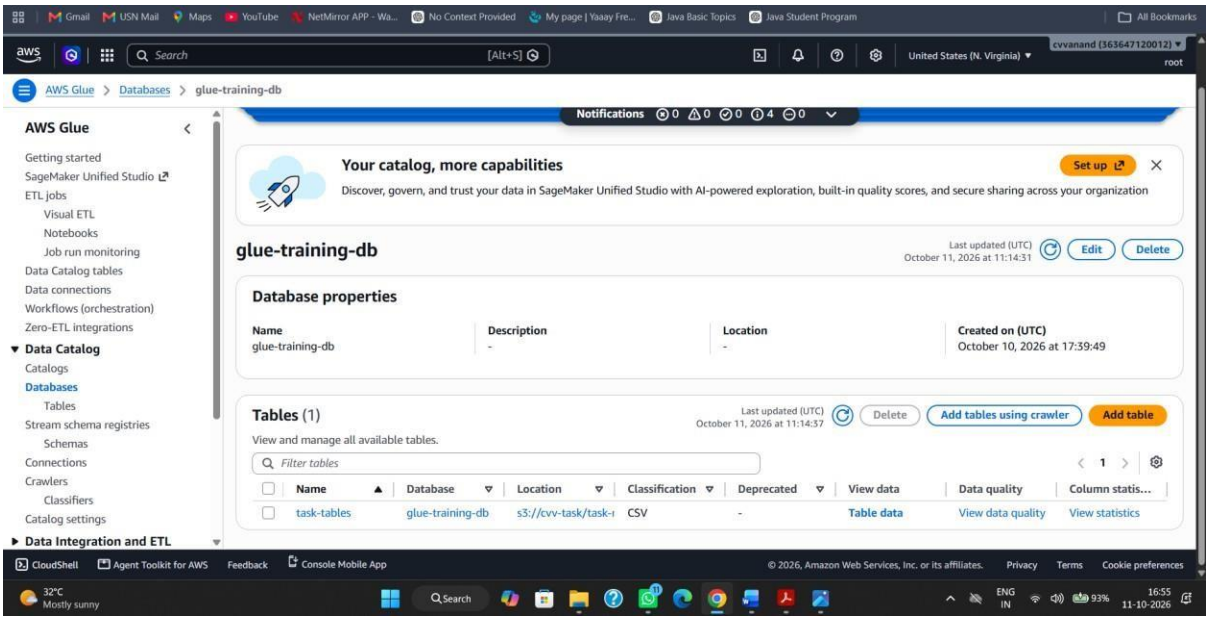


Table successfully created

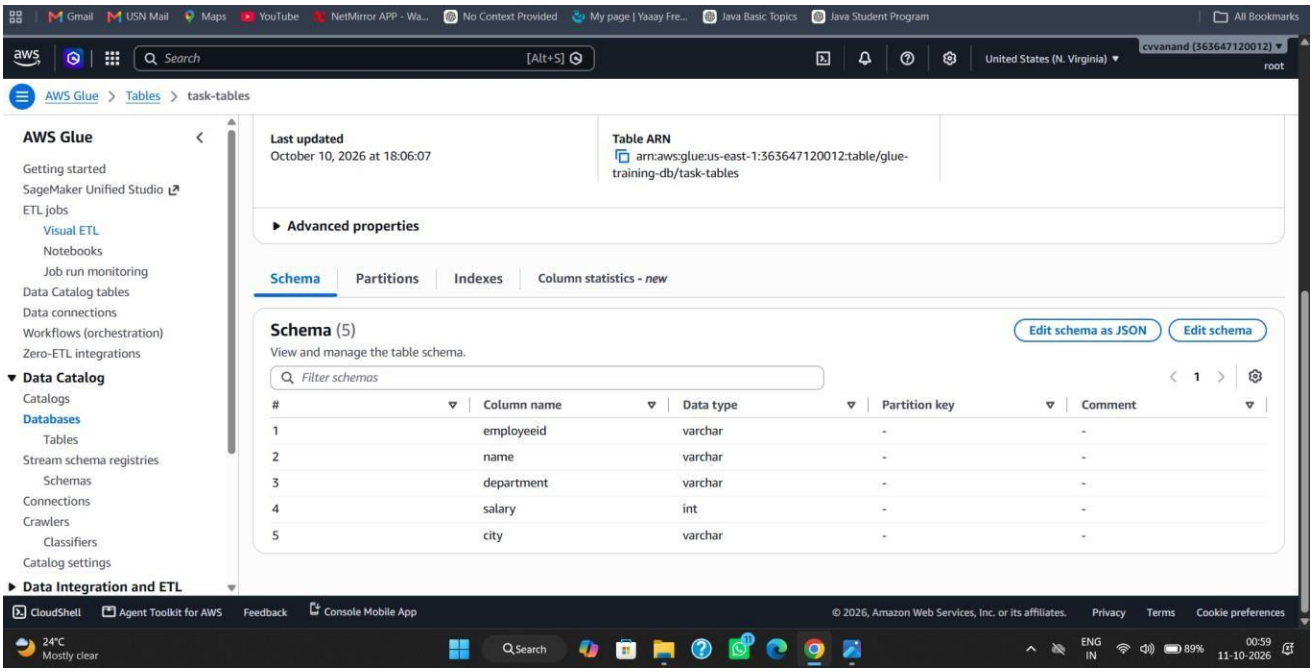
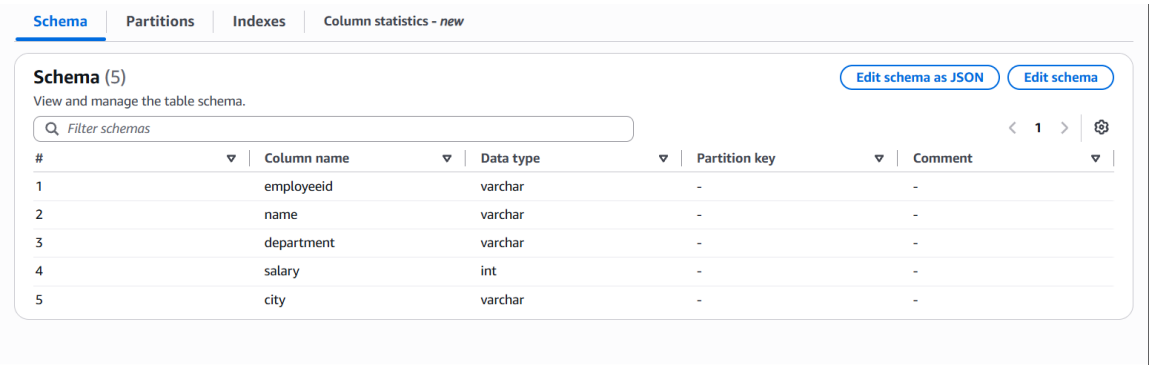


Table Schema: -

employee_id , name, department, salary, city, Data types should also be visible if possible.

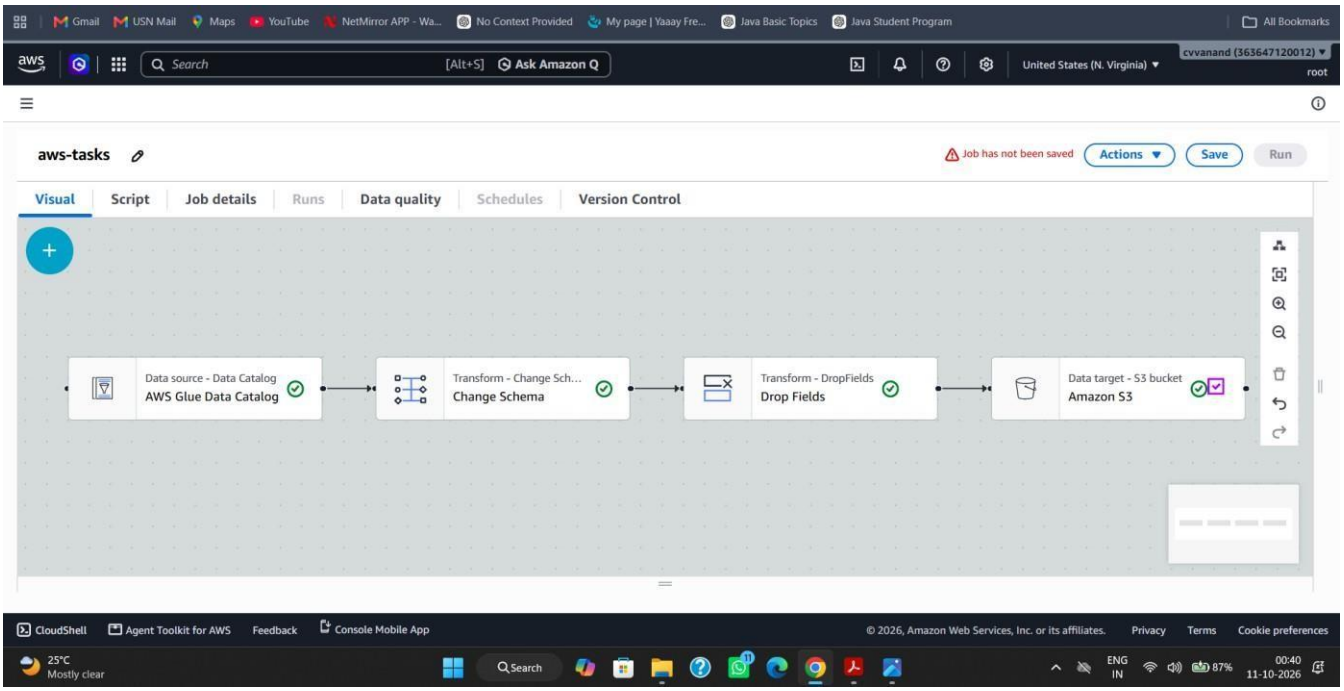


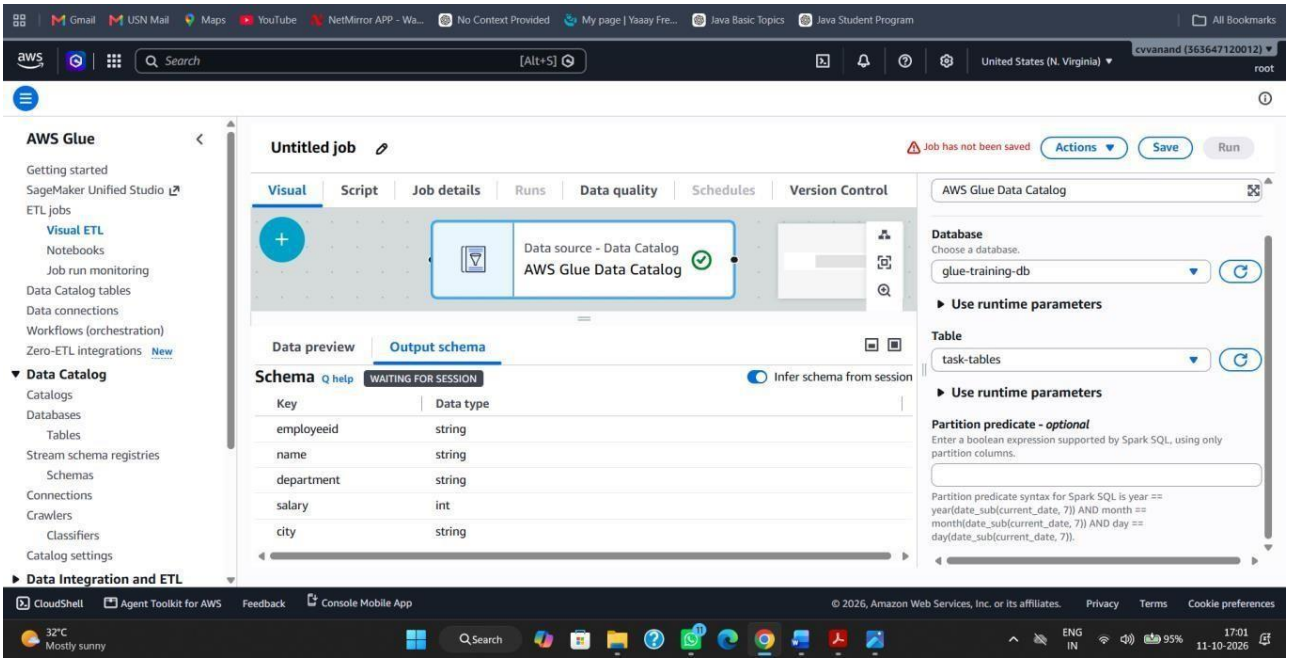
Glue Visual ETL: -

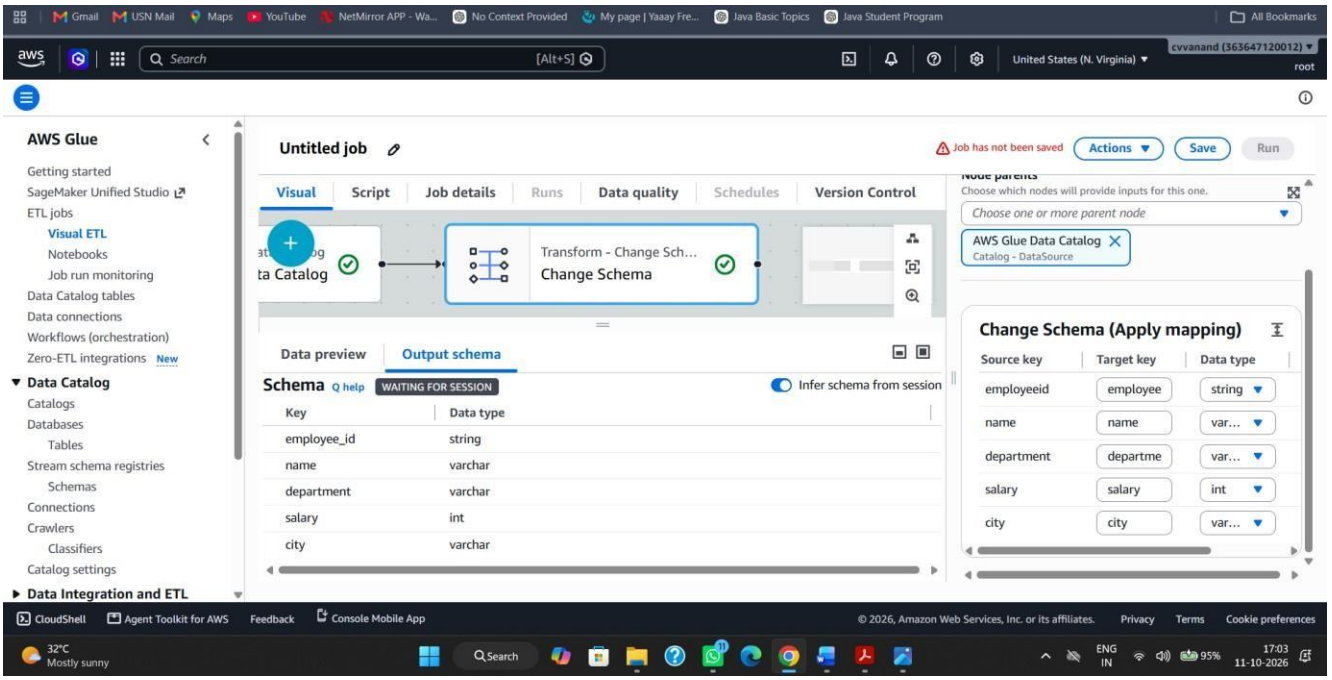
Show the complete workflow:

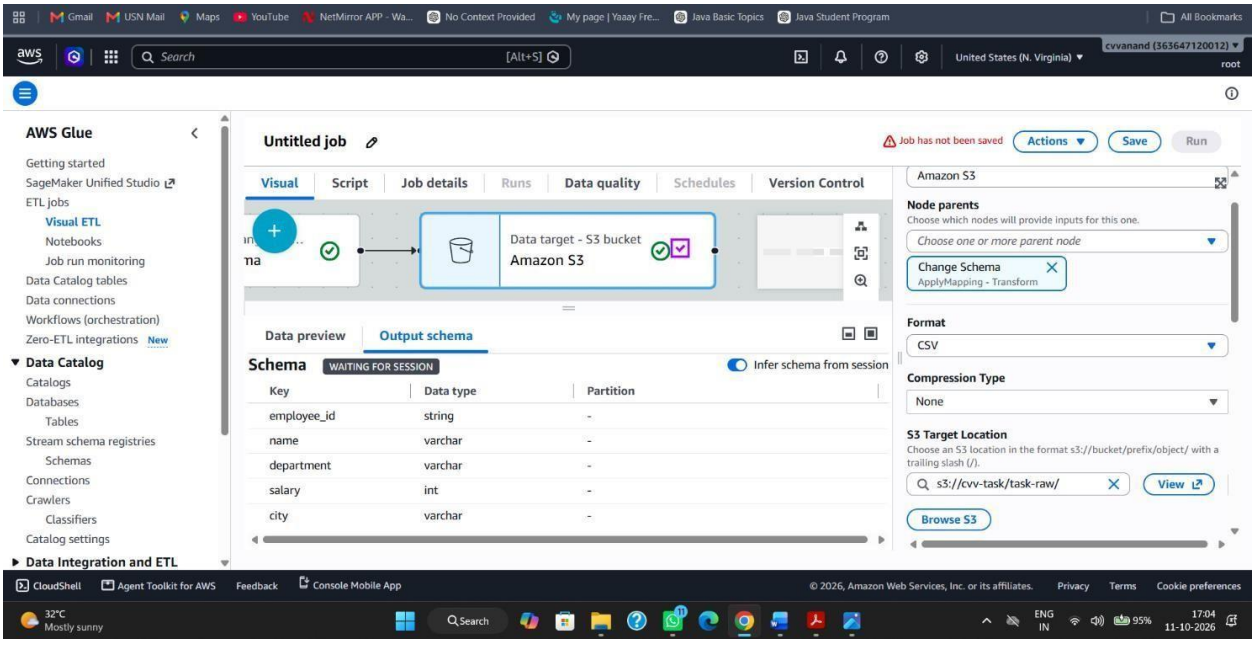
Data Catalog → employees → Transformation → S3 Target

The entire Visual ETL canvas should be visible.






AWS Glue
Untitled job
Job has not been saved
Actions
Save
Run
Visual
Script
Job details
Runs
Data quality
Schedules
Version Control
Transform - Change Sch...
Change Schema
Data preview
Output schema
Schema
WAITING FOR SESSION
Infer schema from session
Key
Data type
employee_id string
name varchar
department varchar
salary int
city varchar
Choose one or more parent node
AWS Glue Data Catalog
Catalog - DataSource
Change Schema (Apply mapping)
Source key
Target key
Data type
employeeid employee string
name name var...
department departme var...
salary salary int
city city var...

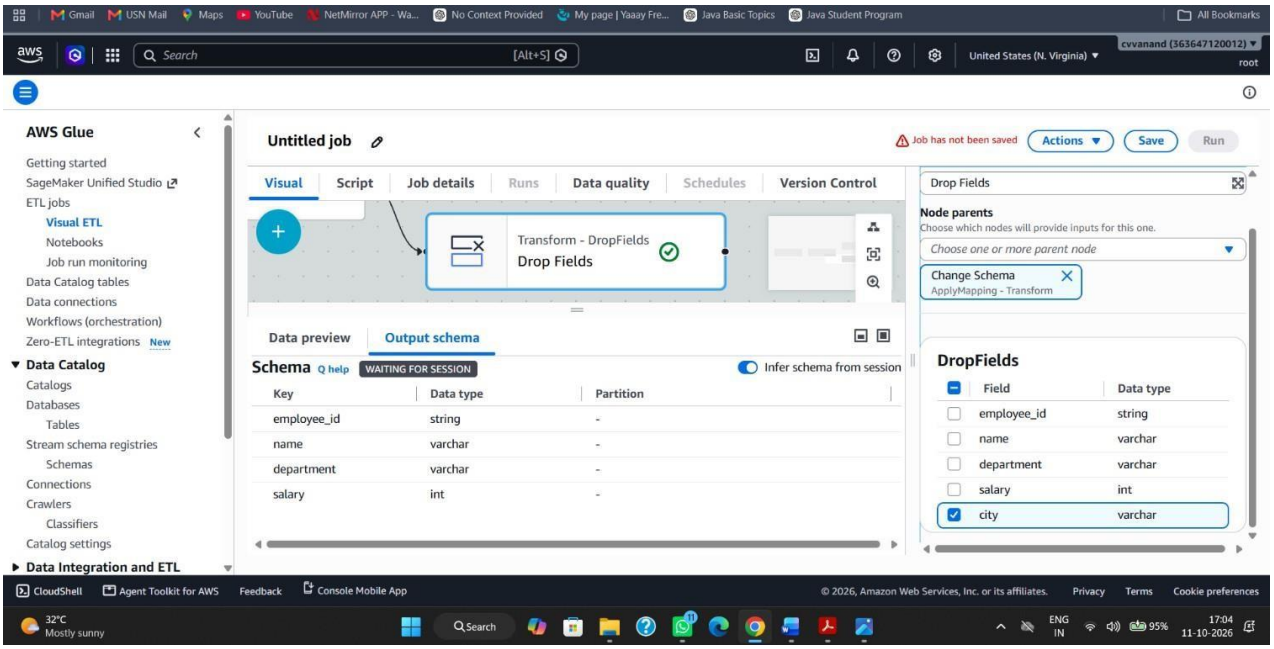

AWS Glue
Untitled job
Job has not been saved
Actions
Save
Run
Visual
Script
Job details
Runs
Data quality
Schedules
Version Control
Data target - S3 bucket
Amazon S3
Data preview
Output schema
Schema
WAITING FOR SESSION
Infer schema from session
Key
Data type
Partition
employee_id string -
name varchar -
department varchar -
salary int -
city varchar -
Amazon S3
Node parents
Choose which nodes will provide inputs for this one.
Choose one or more parent node
Change Schema
ApplyMapping - Transform
Format
CSV
Compression Type
None
S3 Target Location
Choose an S3 location in the format s3://bucket/prefix/object/ with a trailing slash (/).
s3://cvv-task/task-raw/
View
Browse S3

Glue Transformation: -

Show the transformation performed.

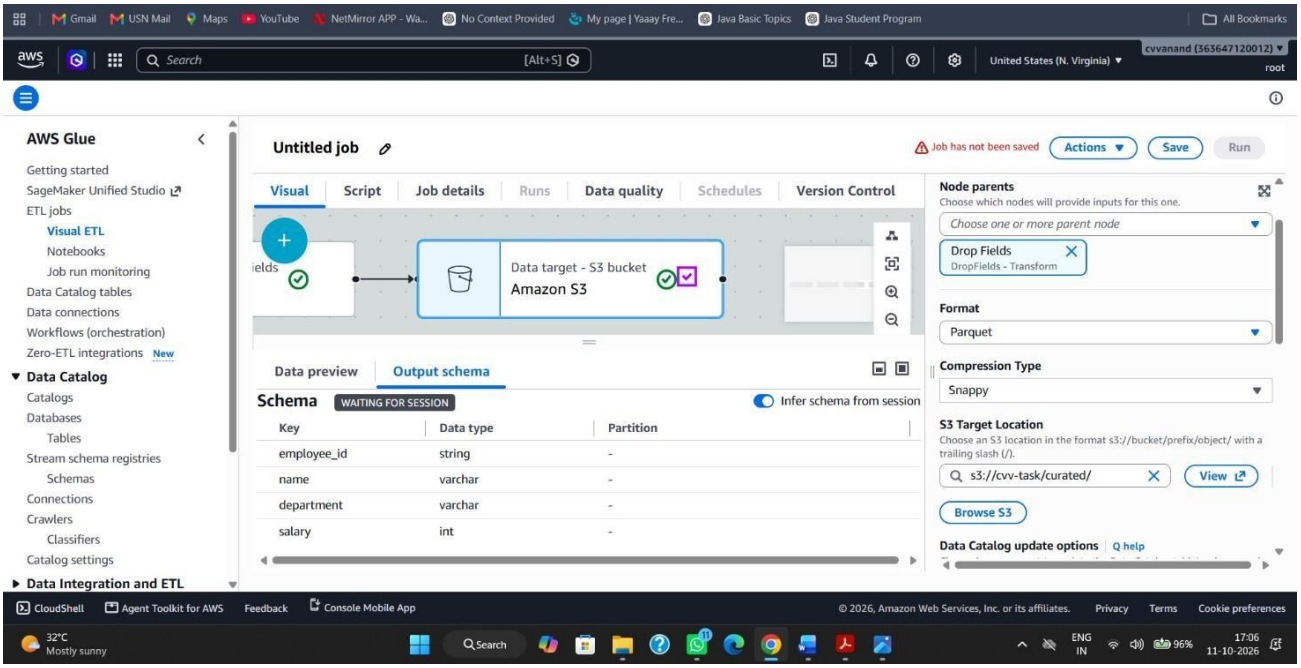
Example: employees → Drop Fields → Remove city

The transformation configuration should be visible.



S3 Target Configuration: -

S3 target, curated/ output location, Output format – Parquet, Relevant target configuration

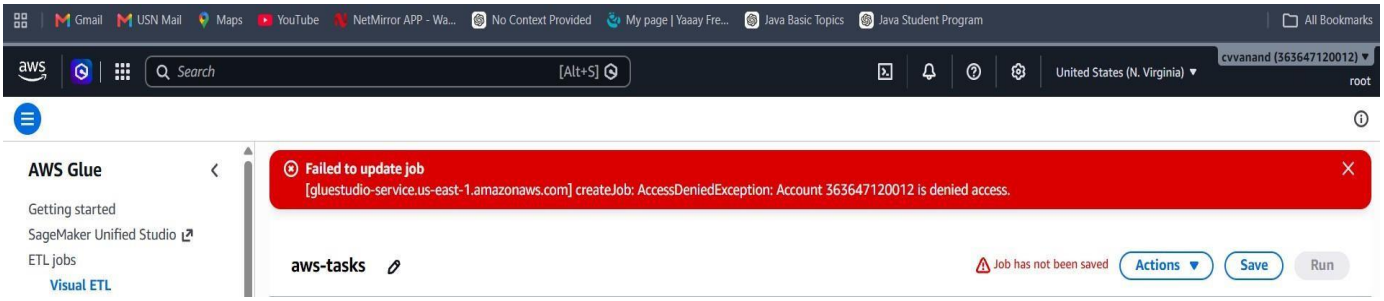


Glue ETL Job Execution: -

Job name, Job status = Succeeded

Final S3 Output: -

curated/ folder, Generated output files, Parquet output



Failed to update job

[gluestudio-service.us-east-1.amazonaws.com]createJob: AccessDeniedException: Account 363647120012 is denied access.